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Revision A

I&C ENVIRONMENTAL MONITORING SPECIFICATION & DESIGN

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REVISION HISTORY

Revision	Date	Description of Changes
A	See Watermark	Initial Release

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ACRONYMS

Acronym	Description
ADC	Analog to Digital Converter
ADC	Analog to Digital Converter
AGC	Automatic Gain Control
AMC	Advance Mezzanine Module
AOM	Acousto-Optical Modulator
CBD	Cabling Diagram
CDA	Clean Dry Air
CO2	Carbon-dioxide laser
CODAC	Control, Data Access and Communication
COTS	Commercial Off-The-Shelf
cRIO	compact Real-time Input-Output
CU	Cubicle (Instrumentation)
CWS	Chilled Water Supply
D*	Detector Sensitivity
DAN	Data Archive Network
DC	Direct Current Voltage
DDS	Direct Digital Synthesis
DIII-D	Tokamak located in San Diego, CA
DPD	Digital Phase Demodulator
DSP	Digital Signal Processing
ENV	TIP Environmental Control Cubicle
EPICS	Experimental Physics and Industrial Control System
FDR	Final Design Review
FMC	FPGA Mezzanine Connector
FPGA	Field-Programmable Gate Array
FSM	Fast Steering Mirror
I&C	Instrumentation & Control
IF	Intermediate Frequency
IQ	In-Phase, Quadrature-Phase
IR	Infra-Red
LAN	Local Area Network

LCC	Local Controller Cubicle
MCB	Miniature Circuit Breakers
MHz	Mega-Herz
OS	Operating System
P&ID	Piping & Instrumentation Diagram
PA	Power Amplifier
PCDH	Plant Systems Design Handbook
PCF	Fast Controller
PCF	Fast Plant Controller
PCS	Plasma Control System
PCSS	Port Cell Support Structure
PDR	Preliminary Design Review
PFD	Process Flow Diagram
PoE	Power Over Ethernet
PON	Plant Operations Network
PSD	Position Sensing Detector (optical)
PSH	Plant System Host
PSM	Piezo Steering Mirror
QCL	Quantum Cascade Laser
QSBA	Quasi-Static Beam Alignment System
R&L	Right & Left Circular Polarization
RF	Radio Frequency
RMS	Root Mean Square
SDN	Synchronous Data Network
SLD	Single Line Diagram
SLO	Synchronous Local Oscillator
SPS	Samples per Second
SST	Stainless Steel
TCN	Timing Control Network
TEC	Thermo-Electric Cooler
TIP	Toroidal Interferometer Polarimeter
TTL	Transistor to Transistor Logic
U	Vertical Unit of Rack Space (1.75")

VAC	Voltage, Alternating Current
VCXO	Voltage Controlled Crystal Oscillator
VGA	Variable Gain Amplifier
VSWR	Voltage Standing Wave Ratio

1 INTRODUCTION

1.1 Purpose

This Document outlines the I&C design of the Environmental Monitoring System, with the objective of defining system architecture and components.

2 SUBSYSTEM OVERVIEW

2.1 Environmental Control System Overview

Environmental Monitoring subsystem consists of temperature sensors mounted along the UWAVS transmission line and optics, including DSM, Interspace, bio-shield, and Back End Optics and Cameras (BEOC) assembly located in the Port Cell.

Thermocouples are mounted on transmission line structure and on select mirrors and optics. The temperatures captured in mirrors are used as thermal background calibration to be applied to images captured on IR cameras, as a part of calibration process. The thermocouples mounted on the structures such as IOT are used to monitor the local temperature and general trends.

All the sensors are monitored by Siemens PLC located in the Diagnostic Hall “Support Cubicle” [R00030]. The cabling diagrams show the connection to thermocouples in detail. [CBD].

2.2 Special Design Considerations

The temperature sensors placed in vessel, interspace, and port cell are subject to high temperatures and radiation exposure. Therefore, sensors in these areas are required to be radiation hardened.

Thermocouples placed in the in-vessel vacuum environment in A10, A20, and A30 sub assemblies must use mineral insulated cables (MIC) due to high radiation and high temperature environment in these modules. Standard thermocouple wire may be used for the remainder of the system outside of vacuum.

2.3 Design Rationalization

The environmental sensors need to be radiation hardened and this automatically limits the selection pool for these sensors and vendors.

ITER has approved the use of N-type thermocouples in the Port Cell and radiation hardened environment. 55.GA will be using N-type thermocouples specified by ITER.

Normally, considering the long distances between deployment on thermocouples in UWAVS beamline to the acquisition in the diagnostic hall Siemens PLC, transmitters would be needed. But due to failure of electronics in the radiation hardened environment, they can’t be used. ITER has performed prototyping to connect the N type thermocouples directly to Siemens PLC Analog module to show the signal integrity and measurement accuracy are not affected over long distance, up to 200m [8MV2HY].

There are 5 UWAVS ports and each port is at different locations and distance from building 74 where diagnostic hall is located. All ports are less than 200m from the diagnostic hall where Siemens PLC is located. The locations and distances are shown in Figure 1 .

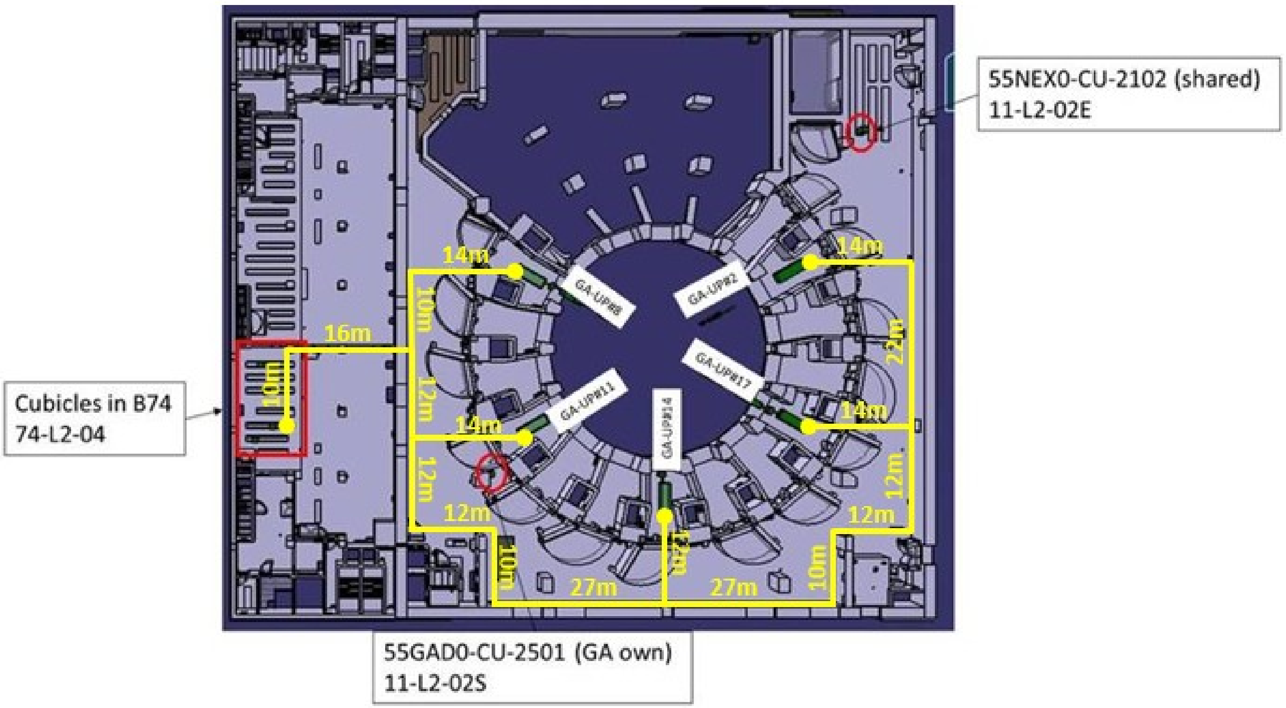

Figure 1. 55.GA cable distances to building 74

3 SYSTEM ARCHITECTURE & DESIGN

3.1 System Architecture

The thermocouples are mounted in structure and optics throughout the UWAVS ports. The location of thermocouples in UWAVS ports and quantity is shown in Table 1. For more details refer to [PID].

Table 1. 55.GA Transmission line thermocouples

Region	UWAVS Component	Thermocouple Quantity
In Vessel	FMU: Aperture	1
	FMU:A10	2
	A20: Shield Module #2	2
	A30: Shield Module #4	1
	A30: Shield Module #5	1
	A30: Shield Module #6	1
	A30: Shield Module #7	1
Interspace	IOT Structure	4

	A40: Assembly M1	3
	A40: Assembly M2	3
	A50:Assembly M1	3
	A50:Assembly M2	3
	A50:Assembly M3	3
	A50:Assembly M4	3
Bio-Shield	A60	6
Port Cell	BEOC	12

There are total of 49 thermocouples that are part of the Environmental Monitoring system. The thermocouples are connected to Siemens PLC S7-1500 during installation / commissioning phase. The data is acquired on PLC continuously after installation of the Environmental Monitoring System. The Siemens PLC S7-1500 is shared between many other UWAVS I&C equipment including thermocouples. The architecture is shown in Figure 2.

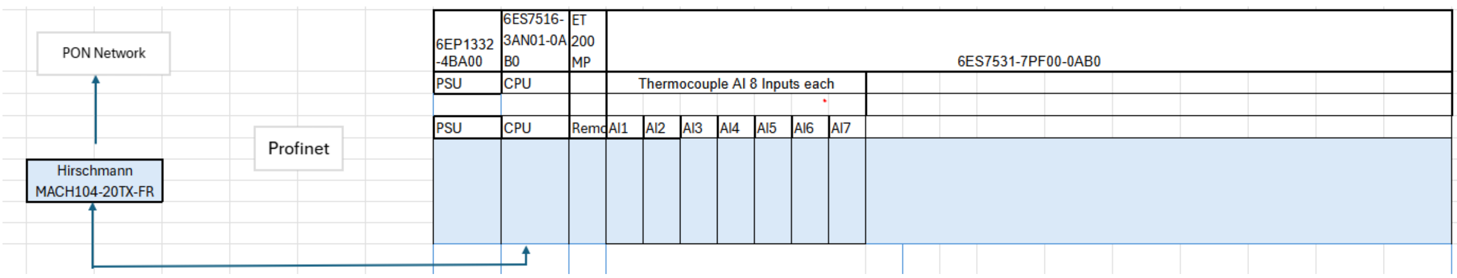

Figure 2. Thermocouple interface with Analog Module and S7-1500 PLC

3.2 Monitoring strategy & Operational Scenarios

The need for environmental sensors dispersed throughout the beamline can be mapped to the following requirements and use cases in Table 2.

Table 2. 55.GA Use cases requirements and functions relating to environmental monitoring

Use case	Requirement	Function
D1-N9-UC-003	D1-N9-MER-BCT-038	GAD0-DME-BCO-FW-ST-I1-MEAS, GAD0-DME-BCO-FW-ST-I2-MEAS
D1-N9-UC-004	D1-N9-MER-MPC-014	GAD0-DME-MPC-TI-EL-ME
D1-N9-UC-005	D1-N9-MER-BCT-032	GAD0-DME-BCO-II-DI-ME, GAD0-DME-BCO-II-TI-ME
D1-N9-UC-006	D1-N9-MER-BCT-032	GAD0-DME-BCO-II-DI-ME, GAD0-DME-BCO-II-TI-ME
D1-N9-UC-007	D1-N9-MER-BCT-053	GAD0-DME-BCO-TI-LH-ME
D1-N9-UC-008	D1-N9-MER-ACT-024	GAD0-DME-ACO-HL-ST-I1-MEAS, GAD0-DME-ACO-HL-ST-I2-MEAS

D1-N9-UC-009	D1-N9-MER-PSD-007	GAD0-DME-PHY-FW-ST-I1-MEAS, GAD0-DME-PHY-FW-ST-I2-MEAS
D1-N9-UC-010	D1-N9-MER-MTR-001	GAD0-DME-OMP-IV-WI-ME
D1-N9-UC-014	D1-N9-MER-PSD-007	GAD0-DME-PHY-FW-ST-I1-MEAS, GAD0-DME-PHY-FW-ST-I2-MEAS

Thermocouples are used for passive monitoring of temperature in different parts of UWAVS port and different optics. The typical monitoring strategy involves setting up the data acquisition rate from thermocouples by interfacing with Siemens PLC S7-1500 and acquiring real time, time stamped data. The thermal analysis data [refer to Mike Simon], is used to set the high/low temperature trigger alarms. There are no interlocks to other UWAVS systems because of alarms but only used for post processing and maintenance purposes.

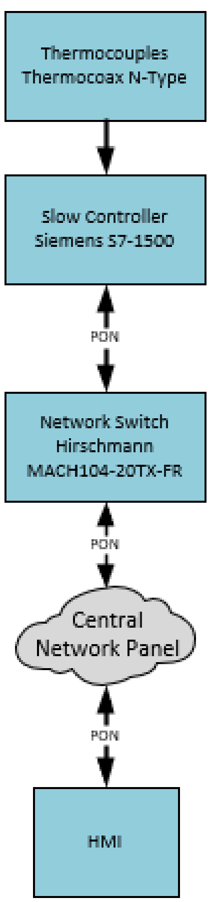

Figure 3. High level view of temperature monitor control scheme

During the commissioning phase the thermocouples are set up with data acquisition rate and temperature is monitored to establish expected response and eliminate faulty thermocouples or interfacing to the PLC.

The historical data for temperature trends for each thermocouple is monitored during the lifetime of ITER to monitor the drift and wear. The thermocouples may be replaced during large maintenance windows based on feedback from UWAVS physicists. Usefulness and need for a drift detection algorithm may be evaluated in the manufacturing phase. Thermocouple data may also be used in future to evaluate failure of cooling lines in A10,A20,A30 UWAVS modules.

The environmental monitoring data is gathered continuously and streamed on CODAC PON after installation at commissioning phase.

Possible exceptions, alarms and error handling is provided in Table .

Table 3. List of possibles errors and alarms

Error Type	Description	Possible Causes	Alarm	Error Handling	Alarm Response
Open Circuit/ Disconnected	Thermocouple input missing or disconnected	Broken wire, loose terminal, thermocouple unplugged	Sensor Fault Alarm	1. Flag for maintenance 2. Use pre determined value as thermocouple input until replaced.	Show "OPEN"
Sensor Shorted	Thermocouple inputs shorted. e.g wire to wire, to grounded surfaces	Pinched wire, installation damage	Sensor Fault Alarm	1. Flag for maintenance 2. Use pre determined value as thermocouple input until replaced.	Show "SHORT"
Out of range reading	Reading outside Thermocouple range (Type N: -40°C to 1300°C)	Faulty thermocouple, signal distortion	Input Range Alarm	1. Flag for maintenance 2. Use pre determined value as thermocouple input until replaced	Show "Invalid Value"
High Temperature Warning	Temperature exceeds defined warning threshold	Normal operating temperature approaching upper limits,	High Temperature Warning	1.Flag for maintenance and physicist during post processing of data.	Show Visual/ audible warning on HMI (highlight temperature/ thermocouple ID)

Overtemperature Fault	Critical temperature limit exceeded	Cooling failure, runaway process		Overtemperature Alarm		Raise alarm on HMI for operator action.
Communication Failure	Link to PLC,CPU,or CODAC is broken	Cable unplugged, CODAC network down		Communication Alarm	1. Flag for maintenance 2. Use pre determined value as thermocouple input until replaced	"No Data" displayed on HMI
Drift or Noise Detected	Fluctuation in readings with constant temperature input	Ground loop, EMI		Sensor Stability Warning	1. Confirm drift or wear using post processing data. 2. Flag for replacement if confirmed.	
Calibration error	Values constantly offset from known standard	Thermocouple degraded/failing		Calibration Warning	1. Confirm drift or wear using post processing data. 2. Flag for replacement if confirmed.	Recommend recalibration/ replacement

As thermocouples are configured at commissioning and gather data continuously, they operate in all plant state operating states (PSOS) specified in the [SDS].

4 TESTING

Commissioning test plan for thermocouples is shown in Table 4.

Table 4. Test Plan

Test ID	**Commissioning 1**
WAVS Area	Diagnostic First Wall(DFW), Diagnostic Shield Module(DSM), Interspace, Bio-shield, Port Cell
Component	Thermocouples
Purpose	Check Functionality of temperature monitoring system
Verification	Document review, test to validate component

Description	•Validate electrical performance, signal accuracy, and functionality of hardware. •Verify measurement stability in room conditions •Verify Data transmission/storage
Support Equipment	Controller units and Cables
Test Conditions	Room Conditions

5 ENVIRONMENTAL MONITORING SYSTEM BOM

Table 5. Environmental Monitor BOM

Device	Manufacturer Number/ ITER Part Number	Quantity
Thermocouple Type N	Thermocoax	49
8 Analog Input Module U/I/RTD/TC	SM531-8AI / 6ES7531-7PF00-0AB0	7
PM 1507 24V/3A power module	6EP1332-4BA00	1
S7-1500	6ES7516-3AN01-0AB0	1

6 OBSOLESCENCE MANAGEMENT

Please refer to ITER_D_PQM7M8 for Obsolescence management on Slow Controller I/O. As UWAVS is using N type thermocouples referred by ITER, UWAVS will reply on obsolescence plan developed by ITER for N type thermocouples.

7 HMI VARIABLES

Table 6. Environmental Monitoring HMI Variables

Variable Type	Variable Name	Description	Typical Format example
Real Time Monitoring	Temp_CH_#	Real time reading from thermocouple in assigned channel	70.0°C
	Status_CH_#	Sensor State	OK, OPEN, SHORT
Alarm and Warning	HI_Temp_CH_#	High temperature Alert, Approaching Upper Limit	TRUE/FALSE
	LO_Temp_CH_#	Low temperature Alert, Approaching Lower Limit	TRUE/FALSE
	CRIT_HI_Temp_CH_#	Critical Temperature Alert, Above Upper Limit	TRUE/FALSE

	CRIT_LO_Temp_CH_#	Critical Temperature Alert, Below Lower Limit	TRUE/FALSE
	Comm_Loss	Loss of Communication with controller/CODAC	TRUE/FALSE
Control and Configuratrion	TEMP_OFFSET_CH_#	Manual adjustment of temperature offset	-1.4°C
	Logging Interval	Manual adjustment between log entries	5 seconds

8 REFERENCES

[2] 55-GA-A0 IR Cameras: VIS/IR TV (Upper) UPP02 PID, ITER_D_PJ4JFG, v2.0.

[3] 55-GA-B0 IR Cameras: VIS/IR TV (Upper) UPP08 PID, ITER_D_PJAC7P, v1.0.

[4] 55-GA-D0 IR Cameras: VIS/IR TV (Upper) UPP14 PID, ITER_D_PHUE63, v1.0.

[5] 55-GA-EO IR Cameras: Vis/IR TV (Upper) UPP14 PID, ITER_D_PJB95T, v1.0.

[6] 55-GA-A0 IR Cameras: VIS/IR (Upper) UPP02 Cabling, ITER_D_3QQP6T, v2.2.

[7] 55-GA-B0 IR Cameras: VIS/IR (Upper) UPP08 Cabling, ITER_D_3QSJ6B, v2.4.

[8] 55-GA-C0 IR Cameras: VIS/IR (Upper) UPP11 Cabling, ITER_D_596WA5, v2.0.

[9] 55-GA-D0 IR Cameras: VIS/IR (Upper) UPP14 Cabling, ITER_D_3QSD6R, v2.0.

[23] 55.GA UWAVS I&C System Design Specification, 304243-S00025, Rev A, ITER_D_ARS734, v1.0.

[25] UWAVS Calibration Plan, 304243-V00057, Rev A, ITER_D_C8HP3D, V1.0.

[30] I&C Cubicle Internal Configuration, ITER_D_4H5DW6, v4.1.

[31] 55.EC-I&C Cubicle Configuration, YSMSJH, V2.1.

[40] 55.GA UWAVS I&C System Requirement Specification, 304243-S00024, Rev A, ITER_D_ARRY56, v1.0.

Please list references.

55.GA - Surface Temperature Measurement and Data Analysis Research and Development Report (UWAV2), ITER_D_YWS8ZC

FBS specification for thermocouple monitoring system, ITER_D_ADAVQH, v1.1

N Type Thermocouple, ITER_D_64J7S9, v1.0

I&C Cubicle Internal Configuration, ITER_D_PQM7M8, v1.0

55.GA_UWAVS-35_System_Detailed_Performance_Definition_rel, 304243-R00018, Rev B